philosophy of this book; however, it is a common method and you need to become familiar with it.

Taking the two-dimensional Fourier transform of the PSF provides the two-dimensional frequency response. If the PSF is circularly symmetric, its frequency response will also be circularly symmetric. In this case, complete information about the frequency response is contained in its profile. That is, after calculating the frequency domain via the FFT method, columns 0 to $N/2$ in row 0 are all that is needed. In imaging jargon, this display of the frequency response is called the **Modulation Transfer Function (MTF)**. Figure 25-1b shows the MTFs for the three PSFs in (a). In cases where the PSF is not circularly symmetric, the entire two-dimensional frequency response contains information. However, it is usually sufficient to know the MTF curves in the vertical and horizontal directions (i.e., columns 0 to $N/2$ in row 0, and rows 0 to $N/2$ in column 0). Take note: this procedure of extracting a row or column from the two-dimensional frequency spectrum is *not* equivalent to taking the one-dimensional FFT of the profiles shown in (a). We will come back to this issue shortly. As shown in Fig. 25-1, similar values of FWHM do not correspond to similar MTF curves.

Figure 25-2 shows a **line pair gauge**, a device used to measure image resolution via the MTF. Line pair gauges come in different forms depending on the particular application. For example, the black and white pattern shown in this figure could be directly used to test video cameras. For an x-ray imaging system, the ribs might be made from lead, with an x-ray transparent material between. The key feature is that the black and white lines have a closer spacing toward one end. When an image is taken of a line pair gauge, the lines at the closely spaced end will be blurred together, while at the other end they will be distinct. Somewhere in the middle the lines will be just barely separable. An observer looks at the image, identifies this location, and reads the corresponding resolution on the calibrated scale.

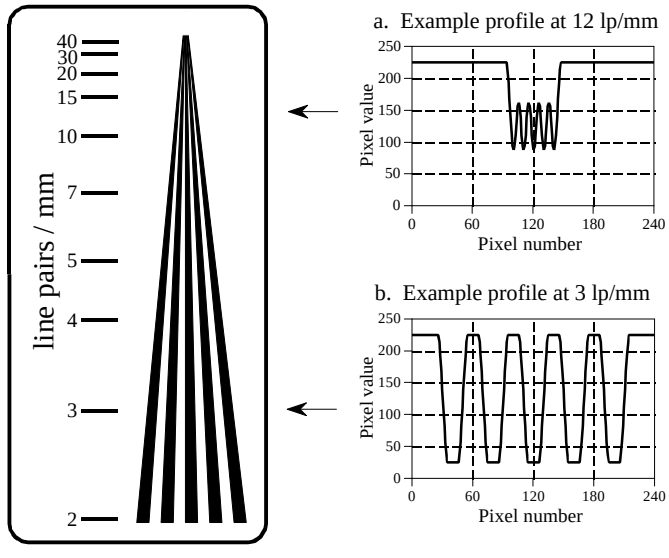

FIGURE 25-2
Line pair gauge. The line pair gauge is a tool used to measure the resolution of imaging systems. A series of black and white ribs move together, creating a continuum of spatial frequencies. The resolution of a system is taken as the frequency where the eye can no longer distinguish the individual ribs. This example line pair gauge is shown several times larger than the calibrated scale indicates.